



Aniruth Ravula

Full Stack Java Developer

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PROFESSIONAL SUMMARY:

- Full Stack Java Developer with 6+ years of experience building enterprise applications across mortgage, banking, financial services.
- Strong experience developing scalable microservices and RESTful APIs using Java 8/11/17/21, Spring Boot, Spring MVC, Spring Security and Hibernate/JPA.
- Experienced in building responsive user interfaces using Angular, React.js, TypeScript, Thymeleaf, jQuery, HTML5 and CSS3.
- Hands-on experience working with SQL and NoSQL databases, including PostgreSQL, Oracle, SQL Server and MongoDB, with expertise in query optimization, indexing and data-access development.
- Experienced in building event-driven, distributed and cloud-based applications using Apache Kafka, Amazon SQS, AWS services, Docker and Terraform.
- Implemented automated CI/CD pipelines using GitHub Actions, Jenkins, AWS CodeBuild and CodePipeline and monitored production applications using CloudWatch, Splunk and AppDynamics.
- Hands-on experience developing machine-learning and deep-learning solutions for text classification, time-series forecasting, image classification and object detection using PyTorch, TensorFlow, Scikit-learn and Statsmodels, with model integration through Flask REST APIs.
- Experienced in performance tuning, root-cause analysis, API validation and code-quality improvement using JProfiler, Chrome DevTools, Postman and SonarQube, following GitFlow and Agile delivery practices.

TECHNICAL SKILLS:

- **Languages and Web:** Core Java 8/11/17/21, JavaScript, TypeScript, Python, Shell/Bash Scripting, Node.js, SQL, PL/SQL, HTML5, CSS3, JSON and XML
- **Frontend:** Angular 12/19, React.js, Thymeleaf, Bootstrap, jQuery and AJAX
- **Backend and APIs:** Spring Boot, Spring MVC, Spring Security, Spring Cloud, Spring AI, Hibernate/JPA, Spring Data JPA, Spring Data MongoDB, JDBC, SQLAlchemy, Flask, Drools, REST, SOAP and Swagger/OpenAPI
- **Databases and Caching:** MySQL, PostgreSQL, Oracle, SQL Server, MongoDB, Azure SQL, Azure Cosmos DB, Amazon ElastiCache for Redis and Memcached
- **Cloud Platforms:** AWS, Azure and GCP
- **AWS Services:** EC2, ECS/Fargate, EKS, ECR, S3, SQS, Lambda, API Gateway, DynamoDB, RDS, EventBridge, ElastiCache, Cognito, Parameter Store, Secrets Manager, CloudWatch, Elastic Beanstalk, CodeBuild, CodePipeline and AWS CLI
- **DevOps and CI/CD:** Git, GitHub, Bitbucket, Docker, Kubernetes, Terraform, Jenkins, Maven, JVM, GitHub Actions, AWS CodePipeline and Azure Pipelines
- **Messaging, Security and Testing:** Kafka, RabbitMQ, JWT, OAuth2, LDAP, JUnit, Mockito, Selenium, Jasmine and Karma
- **Logging, Monitoring and Quality:** SLF4J, Logback, Amazon CloudWatch, Splunk, AppDynamics, Dynatrace, JProfiler, SonarQube, Fortify, Postman and Chrome DevTools
- **Machine Learning:** Scikit-learn, TensorFlow, PyTorch, Statsmodels, XGBoost, LightGBM, Prophet
- **AI-Assisted Development:** GitHub Copilot, Claude, Prodigy AI, Custom AI Skills and Prompt Engineering
- **Tools and Methodologies:** IntelliJ IDEA, VS Code, Eclipse, Jira, Confluence, jBPM, Agile/Scrum

PROFESSIONAL EXPERIENCE:

Xpanse, Fort Lauderdale, FL

Feb 2024 – Present

Full Stack Java Developer

- Designed and developed a cloud-based mortgage rules platform using Java, Spring Boot, Angular, PostgreSQL and AWS services.
- Built scalable Spring Boot microservices and RESTful APIs to support mortgage document processing, rule authoring and business-rule execution workflows.
- Developed backend components using Spring MVC, Spring Data JPA and Hibernate, incorporating request validation, standardized responses and centralized exception handling.
- Developed Spring Boot REST APIs integrated with MongoDB Atlas to store and retrieve loan data, document extraction JSON, and client system inputs for mortgage rule validation.
- Developed Angular 19 standalone components and injectable services using TypeScript, dependency injection, HttpClient and RxJS Observables to integrate with Spring Boot REST APIs for mortgage rule authoring, management and execution.
- Built Angular 19 human-in-the-loop review screens using Reactive Forms, form validation and component communication through @Input and @Output to display document-level rule failures, capture reviewer actions and persist updates through backend APIs.
- Led the migration of more than 2,000 mortgage rules from InRule to a custom Drools-based platform, reducing licensing costs and improving rule maintainability.
- Designed an application-level KieContainer caching mechanism using ConcurrentHashMap, reducing execution time from approximately three seconds to under one second.
- Implemented Amazon ElastiCache for Redis to cache frequently accessed application configuration, reducing downstream calls and improving response time.
- Implemented asynchronous request processing using Amazon SQS to decouple services, support high-volume workloads, and improve fault tolerance.
- Developed Node.js-based AWS Lambda functions integrated with S3, SQS and EventBridge to process rule-engine evaluation responses and store audit records.
- Containerized Spring Boot microservices using Docker, published versioned images to Amazon ECR and deployed services through Amazon ECS.
- Developed automated unit tests using JUnit and Mockito for Spring Boot controllers, services and business-rule components.
- Built and maintained CI/CD pipelines using GitHub Actions for selected repositories and AWS CodeBuild and CodePipeline integrated with GitHub for others, automating testing, SonarQube scanning, Docker image publication to Amazon ECR and deployment to Amazon ECS.
- Implemented structured application logging using SLF4J and Logback across Spring Boot microservices, including appropriate log levels, correlation IDs and exception details, enabling efficient troubleshooting through Amazon CloudWatch.
- Implemented thread-safe caching and concurrent request processing using Java multithreading utilities and ConcurrentHashMap, preventing race conditions while improving the performance of execution.
- Used Dynatrace to monitor the mortgage platform's multiple Spring Boot microservices including document processing, rule authoring, and rule execution services from a centralized dashboard, tracking response times, error rates, service dependencies, and infrastructure metrics to identify performance bottlenecks and troubleshoot production issues.
- Developed data-transformation components to convert incoming MISMO XML mortgage data into JSON for Drools rule evaluation and transform evaluation results from JSON back into MISMO-compliant XML responses for downstream clients.
- Developed reusable Terraform configurations to provision consistent AWS infrastructure across development, QA, UAT and production environments.
- Managed application configuration and feature flags using AWS Systems Manager Parameter Store, enabling controlled functionality across environments.

- Followed Test-Driven Development (TDD) practices by writing JUnit and Mockito tests before implementation, improving code quality and reducing regression issues.
- Optimized PostgreSQL queries and indexes, profiled Java services using JProfiler, and analyzed CloudWatch logs and metrics to resolve application and production-performance issues.
- Integrated Spring AI with Amazon Bedrock APIs to generate Drools Rule Language (DRL) rules from business requirements and validate the generated rules through a dedicated validation API before approval and deployment.
- Leveraged GitHub Copilot to accelerate code development and created a reusable AI skill that translates raw Drools Rule Language (DRL) code into clear, business-friendly English, improving rule understanding and collaboration between technical and business teams.
- Participated in architecture discussions, code reviews, sprint planning and production releases in collaboration with product, QA and DevOps teams.

KLH Engineers, Fort Thomas, KY

Aug 2023 – Dec 2023

Software Engineer

- Developed full-stack applications for MEP engineering workflows using Java, Spring Boot, Angular 13, TypeScript, Azure SQL Database and Azure Blob Storage.
- Built RESTful APIs using Spring Boot, Spring MVC and Hibernate/JPA to manage engineering project data and equipment specifications.
- Developed reusable Angular components and services and integrated them with Spring Boot APIs for interactive engineering dashboards.
- Implemented authentication and authorization using Spring Security and OAuth2 to secure application endpoints and engineering data.
- Integrated Azure SQL Database for structured application data and Azure Blob Storage for storing and retrieving application files.
- Documented and tested REST APIs using Swagger/OpenAPI and Postman to support frontend and backend integration.
- Engineered time-series forecasting models to predict future revenue and incoming MEP project volume, supporting project planning and resource-allocation decisions.
- Evaluated SARIMAX, Prophet, XGBoost, LightGBM, N-BEATS and LSTM models to identify the most accurate forecasting approach.
- Developed an NLP-based naming-standardization solution using LSTM, GRU and BERT to convert CAD/Revit labels into KLH-specific naming conventions, reducing manual effort by 80%.
- Evaluated NLP models using accuracy, precision, recall, F1-score, and confusion matrices to select the most reliable classification approach.
- Built Flask REST API endpoints to expose forecasting and NLP models and integrated them with Autodesk Revit, allowing engineers to access predictions and naming recommendations within their existing workflows.

Tata Consultancy Services / Citibanamex, Chennai, IN

Nov 2020 – Jul 2022

Full Stack Java Developer

- Engineered a monitoring platform using Java, Spring Boot, React.js, Thymeleaf, jQuery, and MongoDB to visualize metrics obtained from Splunk and AppDynamics.
- Implemented a five-second polling service that retrieved and transformed monitoring data from Splunk and AppDynamics and persisted in MongoDB.
- Developed REST APIs, MongoDB repositories and operational dashboards with filtering and Excel-export capabilities.
- Developed an ARIMA-based proof of concept to forecast server CPU and memory utilization from historical metrics; following successful validation, the forecasting capability was adopted for application-wide infrastructure monitoring.
- Implemented Kafka-based asynchronous communication between Spring Boot microservices to decouple banking workflows and support reliable message processing during peak workloads.
- Deployed Docker-containerized Spring Boot microservices on Kubernetes using Amazon EKS, with AWS CodeBuild and AWS CodePipeline automating build and deployment workflows.

- Integrated Spring Boot services with MongoDB using Spring Data MongoDB for operational data persistence and retrieval.
- Developed retry mechanisms and centralized exception-handling components to provide consistent error responses and improve service recovery across distributed applications.
- Configured cron jobs to invoke application API endpoints every three minutes for scheduled data processing.
- Created complex PL/SQL procedures, optimized database queries, and developed analytical views in Oracle and SQL Server for compliance and reporting requirements.
- Maintained Jenkins CI/CD pipelines with SonarQube and Fortify quality gates and supported deployments and rollback procedures across multiple environments.

Bank of America, Hyderabad, IN

Apr 2019 – Oct 2020

Full Stack Java Developer

- Developed an internal reconciliation platform using Java, Spring Boot, Angular and TypeScript for Bank of America financial analysts.
- Built Spring Boot REST APIs to retrieve and compare account and portfolio data from multiple systems, identifying missing, inconsistent and mismatched records.
- Developed Angular dashboards to present reconciliation results and portfolio differences for analyst review.
- Implemented authentication and role-based authorization using Amazon Cognito, LDAP and OAuth 2.0 to protect financial data and application functionality.
- Created stored procedures and database views and optimized SQL queries in Amazon RDS PostgreSQL to process reconciliation results and generate audit and regulatory reports.
- Implemented Amazon ElastiCache for Redis to cache frequently accessed reconciliation data, reducing database calls and improving application response time.
- Containerized Spring Boot applications using Docker and deployed them to Amazon ECS Fargate.
- Managed application secrets and environment-specific configuration using AWS Secrets Manager and Systems Manager Parameter Store.
- Automated builds and deployments using AWS CodePipeline and CodeBuild and monitored application health, logs and metrics through Amazon CloudWatch.
- Developed unit and integration tests using JUnit, Mockito and Selenium to validate backend services and end-to-end reconciliation workflows.

Certifications:

- Azure Fundamentals – Microsoft
- Splunk 7.x Fundamentals Part 1 – Splunk
- Time Series Analysis – Kaggle
- Associate Cloud Engineer – Google Cloud Platform
- Learning Linux Shell Scripting – LinkedIn Learning

Education:

Master of Engineering in Artificial Intelligence, University of Cincinnati, USA